



Source: Wikimedia

An array of different robotic sensors

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey orders given to it by human beings except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

The implications of these laws to a robotic brain are that it must absolutely always be a top priority to protect the life of any identifiable human being. However, this begins to enter shaky territory. What if the robot is simply unable to discern that a human being stands in danger? Several automated production lines lack any kind of check to see if a person has fallen in - it is expected that the vigilance of the person running or supervising the machine will take care of this. Thus, in order for a machine to completely protect human life, it must be at all times aware of all the humans in its vicinity - this may not always be practical.

The next law becomes problematic the moment we give robots the agency to actually take a human life, even in indirect ways. The recent crisis with the Tesla driver who was killed while using autopilot mode on a driverless car prototype stands as a stark example. Clearly, if we are to make robots capable of relatively dangerous tasks like driving cars and piloting planes, there is a the chance that a certain reordering of priorities would make it acceptable to these robots to take human lives. Similarly, the use of unmanned aerial drones for surveillance and warfare in the Middle East has raised serious concerns about the ability of a completely autonomous object to identify and destroy entire villages.